CASE STUDY

β-Phosphoglucomutase: a 2D QC/MM surface

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run with the 3.2.0 development version.

β-Phosphoglucomutase (β-PGM) converts β-glucose 1-phosphate into glucose 6-phosphate through β-glucose 1,6-bisphosphate (G16P). In the step studied here the phosphoryl group on O1 of G16P is transferred to **Asp8**, the enzyme nucleophile, while **Asp10**, the general acid, gives a proton to the leaving O1:

G16P + $Asp8^{-}$ + Asp10–COOH → glucose 6-phosphate + $Asp8–PO_3^{2-}$ + $Asp10^{-}$

Two events, two coordinates: are they concerted, or does one come first? A 2D scan answers it.

The system

`examples/tutorials/case_studies/5_pgmase/phosphoglucomutase_AMBER_AM1dPhoT.pkl` in the EasyHybrid folder:

Model	β-PGM with G16P and the catalytic Mg^{2+}, pruned around the active site
Force field	AMBER
Atoms	9395; 2806 free atoms
QC region	64 atoms: G16P, Asp10 (protonated), the side chains of Asp8, Asp170 and Lys145, Mg^{2+} and two water molecules, closed by 6 link atoms; **AM1/d-PhoT** (AM1/d reparametrized for phosphoryl transfer), charge −1

The reaction coordinates

Both are **multiple distance** coordinates with the transferred atom in the middle:

	Atoms (#1, #2, #3)	Indices	Start	Meaning
RC1	O1 (G16P), P′ (G16P), OD1 (Asp8)	3439, 3449, 134	−1.64 Å	d(O1–P′) − d(P′–OD1): the phosphoryl leaves O1 for Asp8
RC2	OD2 (Asp10), HD2 (Asp10), O1 (G16P)	166, 167, 3439	−1.75 Å	d(OD2–HD2) − d(HD2–O1): the proton leaves Asp10 for O1

For each coordinate pick its three atoms in **Picking** mode and click **Import from Picking Selection** in its box. The end atoms of both coordinates are oxygens, so the restraints are not mass weighted.

The scan

Simulate → Reaction Coordinate Scans, with **Reaction Coordinate 2** ticked:

	RC1	RC2
Step Size	0.2 Å	0.15 Å
Number of Steps	15	20
Force Constant	4000 kJ mol^{-1} $Å^{-2}$	4000 kJ mol^{-1} $Å^{-2}$

Method Conjugate Gradient, **Maximum iterations** 6000, **RMS Grad. Tolerance** 0.1, **Number of CPUs** 10, **Trajectory Name** `PGM_scan2D`.

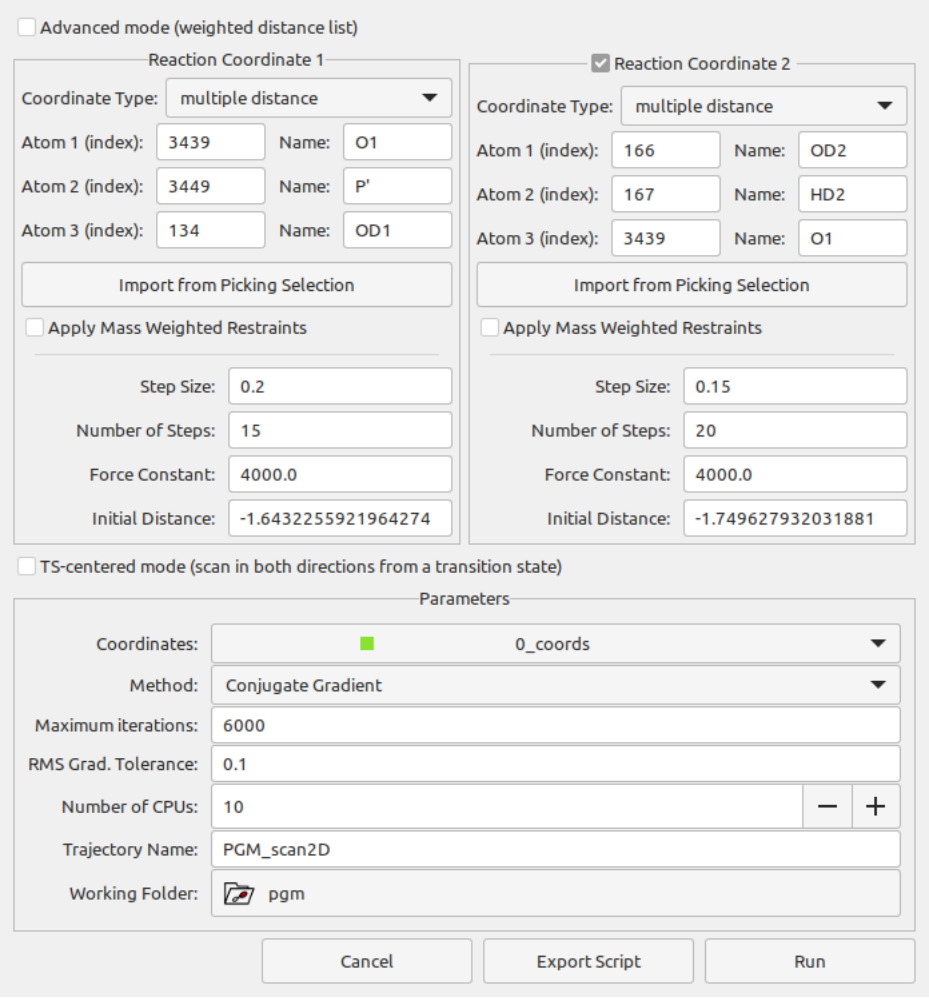


Figure 1. The two coordinates of the 2D scan.

The 300 points take about 18 minutes on 10 cores (each energy costs 0.07 s). Import the scan from the *Finished!* dialog (format *pkl folder 2D*) and open **Analysis → PES Analysis**.

The surface

In the map, rows are RC1 (phosphoryl transfer, from top to bottom) and columns RC2 (proton transfer, from left to right). Tick **Mark minima and saddle points** (Figure 2).

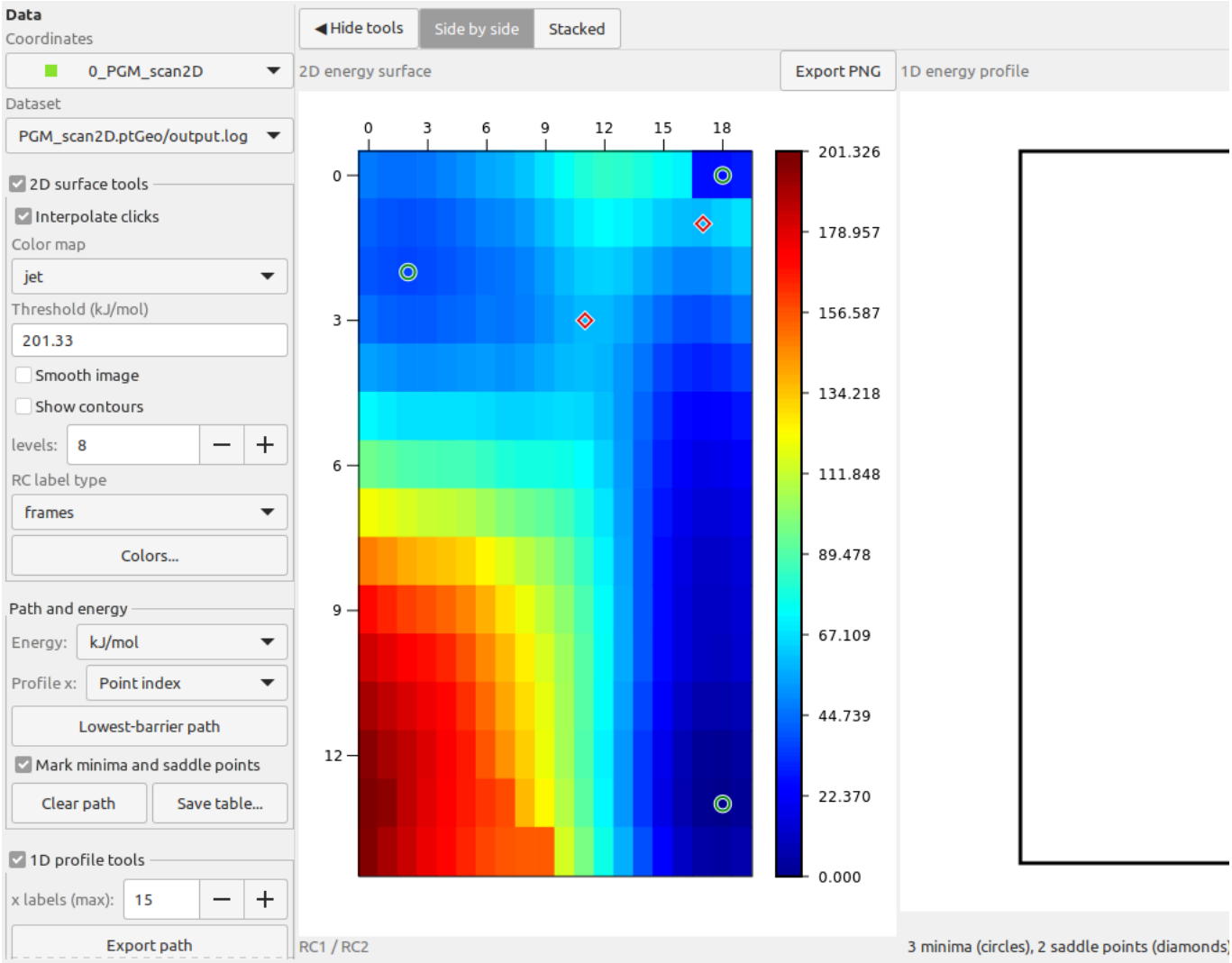


Figure 2. AM1/d-PhoT/AMBER surface (kJ/mol, relative to the lowest point).

- The **reactant** basin is near the top left corner (RC1 = −1.24, RC2 = −1.45 Å; 36.9 kJ/mol).
- The **product**, the phosphoryl on Asp8 and the proton on O1, is the global minimum at the bottom right (RC1 = 0.96, RC2 = 0.95 Å).
- The bottom left corner, phosphoryl transferred with the proton still on Asp10, is the highest region: the phosphoryl group does not leave an unprotonated O1.

Choose the end points yourself. Pressed with no points, **Lowest-barrier path** goes from the global minimum to the lowest other minimum it finds, here the cells at the right end of the first row. Those three points (RC1 = −1.64 Å, RC2 = 0.80–1.10 Å) are 30 kJ/mol below their neighbours: their optimizations fell into a different arrangement of the active site, a discontinuity of the scan, not a state of the reaction. Click the reactant cell and the product cell on the map and then press **Lowest-barrier path** (Figure 3).

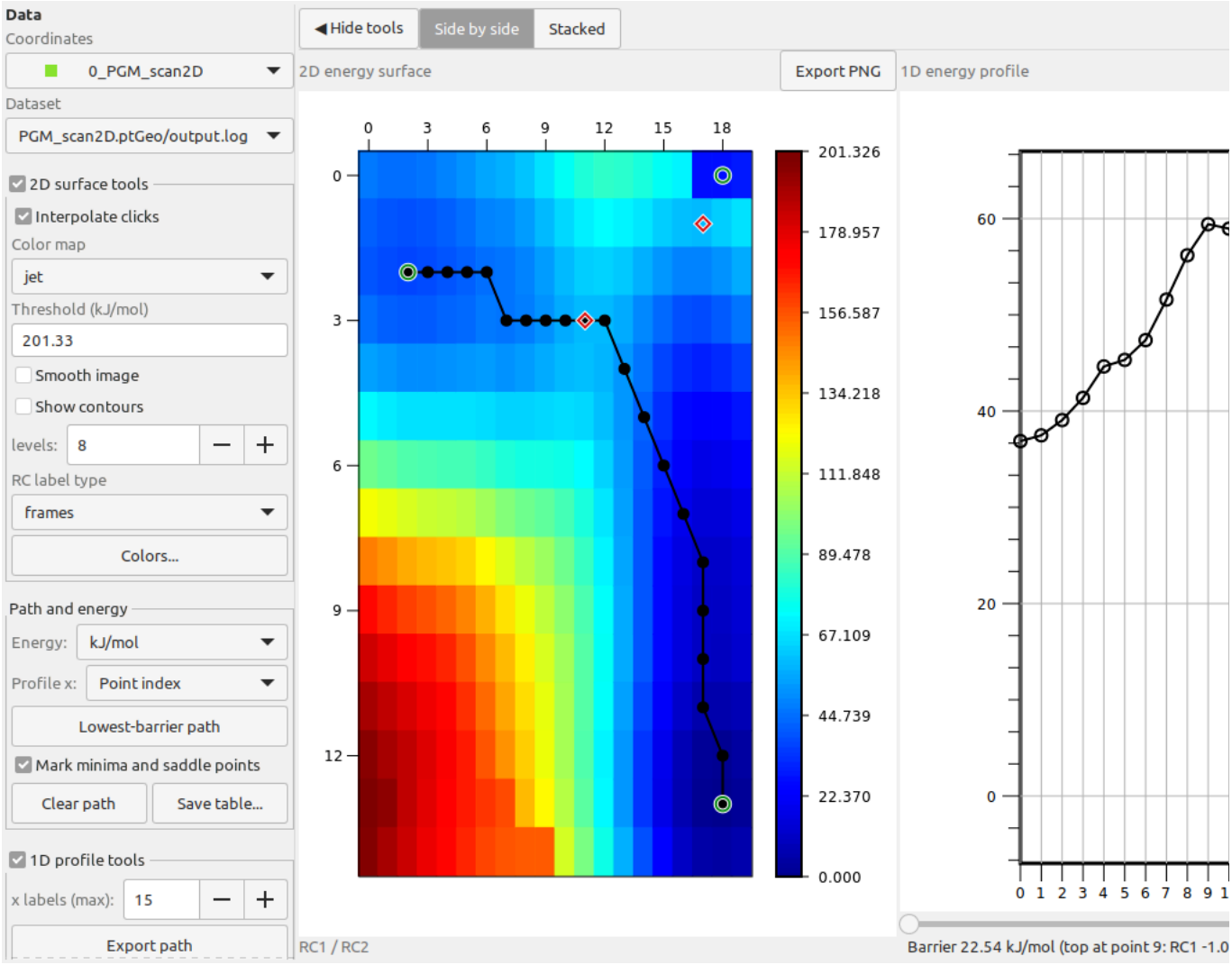


Figure 3. The lowest-barrier path from the reactant basin to the product.

	RC1 (Å)	RC2 (Å)	E − E(reactant) (kJ/mol)
Reactant	−1.24	−1.45	0.0
Highest point of the path	−1.04	−0.10	**22.5**
Product	0.96	0.95	−36.9

Mechanism. Along the path the proton moves first (to the right on the map): RC2 goes from −1.45 to −0.10 Å (the proton halfway between Asp10 and O1) while RC1 changes by only 0.2 Å. After the top the phosphoryl group moves to Asp8 and the proton completes its transfer, downhill. The general acid prepares the leaving group before the P–O1 bond breaks: an asynchronous, concerted mechanism on this surface.

Click a cell to show its structure; the reactant, the top of the path and the product are shown in Figures 4–6.

	O1–P′	P′–OD1 (Asp8)	OD2–HD2 (Asp10)	HD2–O1
Reactant	1.70	2.94	1.05	2.50
Top of the path	1.81	2.86	1.21	1.31
Product	2.78	1.82	1.95	1.00

(distances in Å)

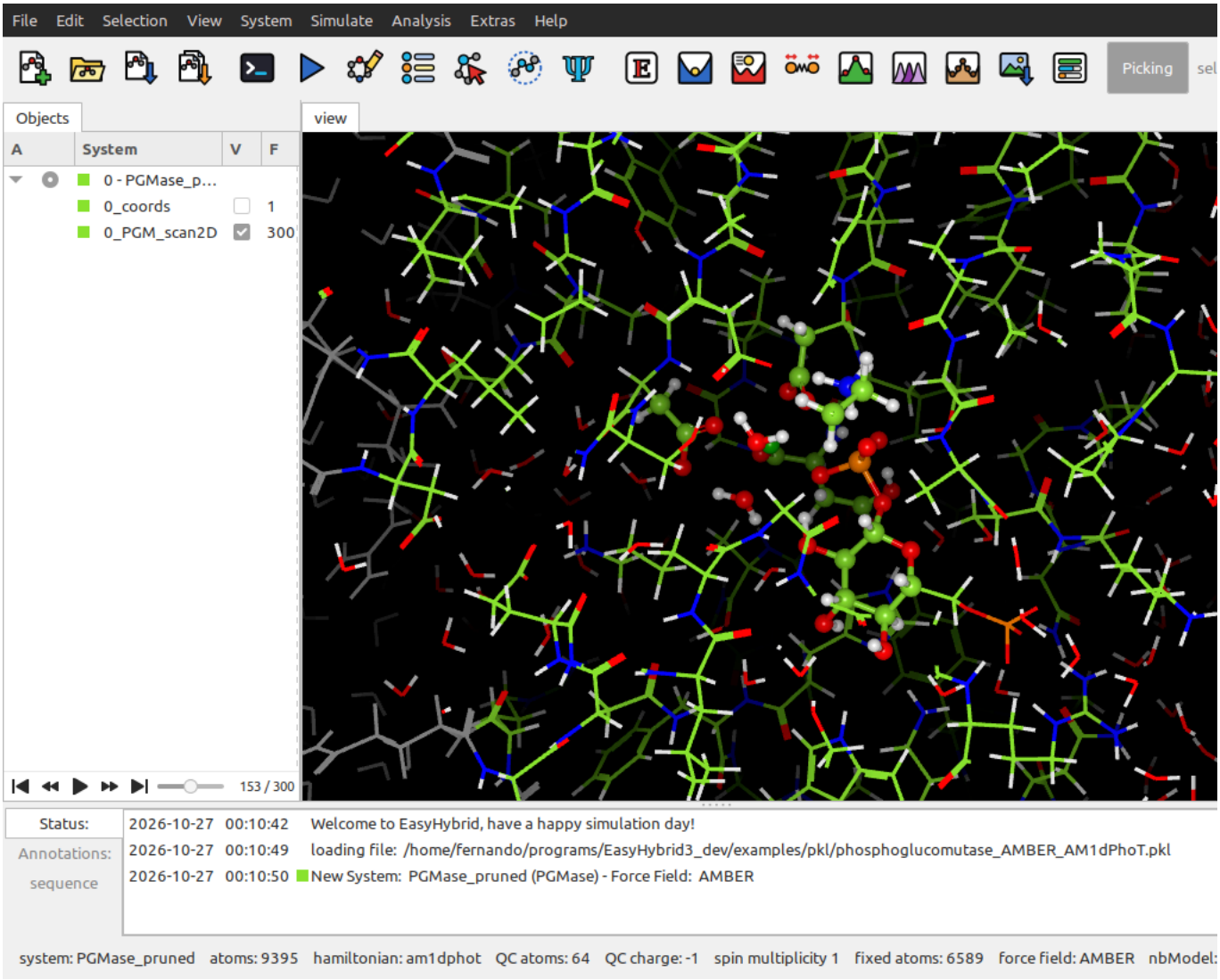


Figure 4. Reactant.

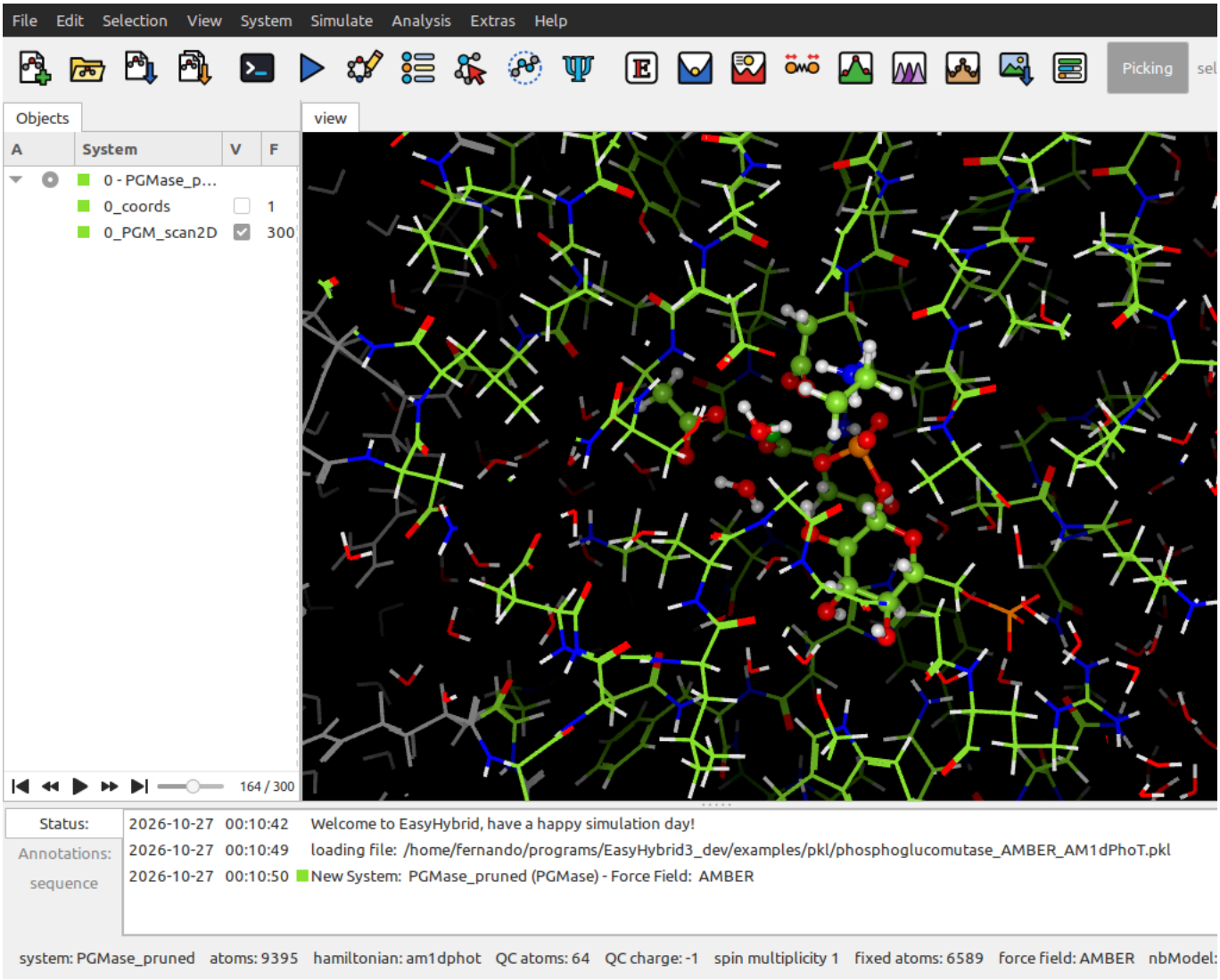


Figure 5. The highest point of the path: the proton between Asp10 and O1.

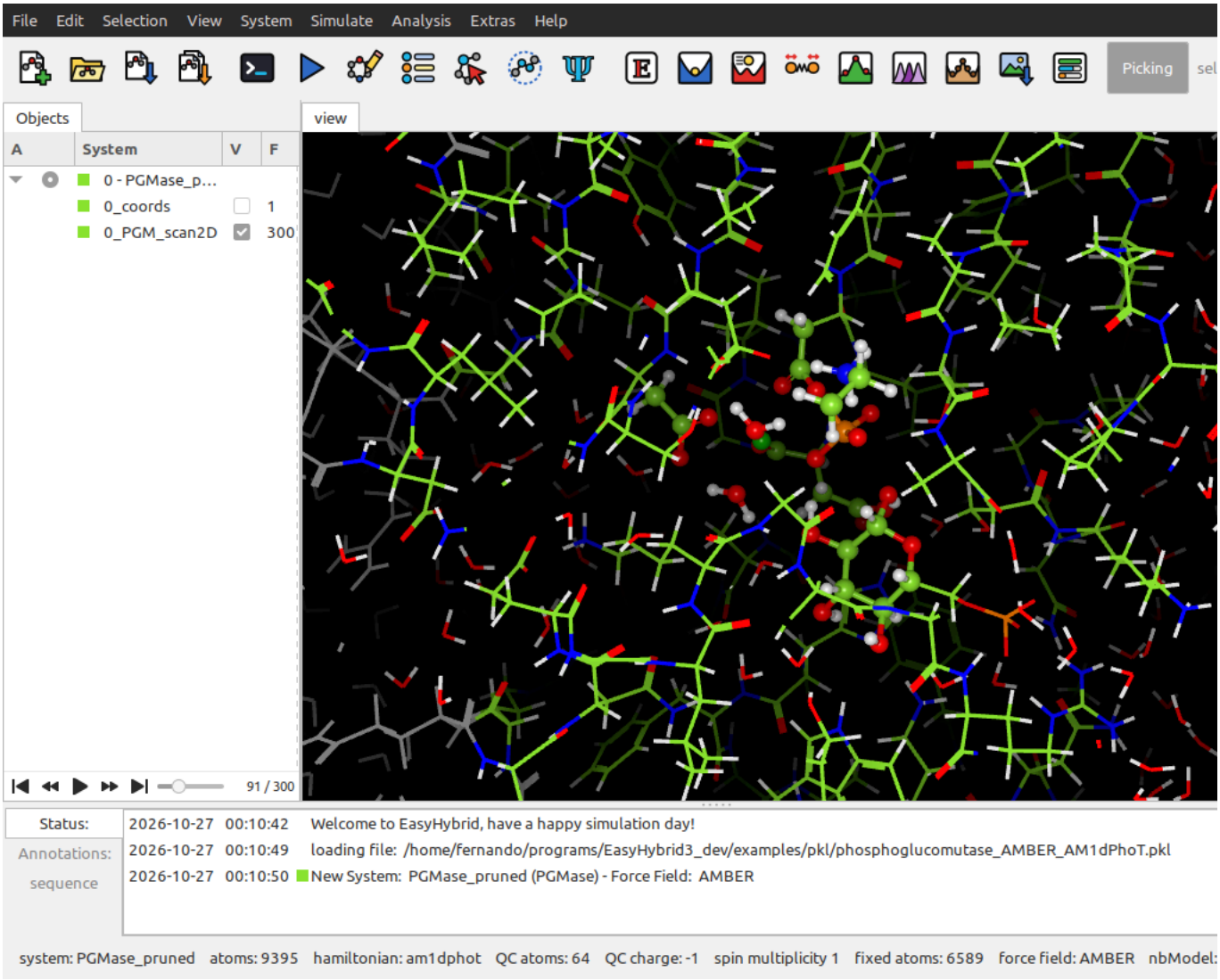


Figure 6. Product: the phosphoryl on Asp8.

Notes

- **Grid resolution.** With 0.15–0.2 Å steps the top of the path is known within a cell; refine it with a finer scan around it or with **Simulate → Search Transition State (Baker)**.
- **Discontinuities.** Large relaxed scans of enzymes always have some. Spot them on the map (isolated cells much lower or higher than their neighbours) and do not let them define the path.
- **Energy refinement.** AM1/d-PhoT is fitted to phosphoryl transfer; **Analysis → Energy Refinement** recomputes the energies of all 300 structures with another method (for example DFT with ORCA) for a better surface.
- References: L. W. Tremblay, G. Zhang, J. Dai, D. Dunaway-Mariano, K. N. Allen, *J. Am. Chem. Soc.* **2005**, 127, 5298 (β-PGM); K. Nam, Q. Cui, J. Gao, D. M. York, *J. Chem. Theory Comput.* **2007**, 3, 486 (AM1/d-PhoT).